Common/rare choice bias relative to non-naive neutral trials — Exact session ISI extremes

All subjects — pooled — all blocks

All subjects — pooled — short block

All subjects — pooled — long block

All subjects — grand average — all blocks

All subjects — grand average — short block

All subjects — grand average — long block

MC06 — pooled — all blocks

MC06 — pooled — short block

MC06 — pooled — long block

MC06 — grand average — all blocks

MC06 — grand average — short block

MC06 — grand average — long block

Block versus matched non-naive neutral P(Choose Long-Associated Response) — Exact session ISI extremes
Pooled: Fisher exact; grand average: paired Wilcoxon

All subjects — pooled — all blocks

All subjects — pooled — short block

All subjects — pooled — long block

All subjects — grand average — all blocks

All subjects — grand average — short block

All subjects — grand average — long block

MC06 — pooled — all blocks

MC06 — pooled — short block

MC06 — pooled — long block

MC06 — grand average — all blocks

MC06 — grand average — short block

MC06 — grand average — long block

Bias distributions — Exact session ISI extremes: median + IQR and two-sided tests versus zero
All visual summaries use median + IQR; Shapiro p < 0.01 selects Wilcoxon, otherwise one-sample t-test

All subjects — pooled — all blocks
All subjects — pooled — short block
All subjects — pooled — long block
All subjects — grand average — all blocks
All subjects — grand average — short block
All subjects — grand average — long block
MC06 — pooled — all blocks
MC06 — pooled — short block
MC06 — pooled — long block
MC06 — grand average — all blocks
MC06 — grand average — short block
MC06 — grand average — long block
Bias distribution
ΔP(Long-associated response)
−0.2
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.8
1.0
ns
Common easy
Common difficult
Rare easy
Rare difficult
n=4
n=2
one-sample t

Common/rare choice bias relative to non-naive neutral trials — Four psychometric bins (outer bins define easy/hard)
Choice bias: ΔP(Choose Long-Associated Response) (block − matched neutral)
○ mean ± SEM
All subjects — pooled — all blocks
0.4
0.2
0.0
−0.2
−0.4
ns
*
ns
ns
Common easy
2 sess 110 trials
Common difficult
2 sess 93 trials
Rare easy
2 sess 16 trials
Rare difficult
2 sess 29 trials
All subjects — pooled — short block
0.6
0.4
0.2
0.0
−0.2
−0.4
ns
ns
ns
ns
Common easy
2 sess 53 trials
Common difficult
2 sess 47 trials
Rare easy
2 sess 5 trials
Rare difficult
2 sess 19 trials
All subjects — pooled — long block
0.8
0.6
0.4
0.2
0.0
−0.2
ns
**
ns
ns
Common easy
2 sess 57 trials
Common difficult
2 sess 46 trials
Rare easy
2 sess 11 trials
Rare difficult
2 sess 10 trials
All subjects — grand average — all blocks
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
ns
ns
ns
ns
Common easy
2 sess 110 trials
Common difficult
2 sess 93 trials
Rare easy
2 sess 16 trials
Rare difficult
2 sess 29 trials
All subjects — grand average — short block
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
ns
ns
n/a
ns
Common easy
2 sess 53 trials
Common difficult
2 sess 47 trials
Rare easy
2 sess 5 trials
Rare difficult
2 sess 19 trials
All subjects — grand average — long block
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
ns
ns
ns
ns
Common easy
2 sess 57 trials
Common difficult
2 sess 46 trials
Rare easy
2 sess 11 trials
Rare difficult
2 sess 10 trials
MC06 — pooled — all blocks
0.4
0.2
0.0
−0.2
−0.4
ns
*
ns
ns
Common easy
2 sess 110 trials
Common difficult
2 sess 93 trials
Rare easy
2 sess 16 trials
Rare difficult
2 sess 29 trials
MC06 — pooled — short block
0.6
0.4
0.2
0.0
−0.2
−0.4
ns
ns
ns
ns
Common easy
2 sess 53 trials
Common difficult
2 sess 47 trials
Rare easy
2 sess 5 trials
Rare difficult
2 sess 19 trials
MC06 — pooled — long block
0.8
0.6
0.4
0.2
0.0
−0.2
ns
**
ns
ns
Common easy
2 sess 57 trials
Common difficult
2 sess 46 trials
Rare easy
2 sess 11 trials
Rare difficult
2 sess 10 trials
MC06 — grand average — all blocks
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
ns
ns
ns
ns
Common easy
2 sess 110 trials
Common difficult
2 sess 93 trials
Rare easy
2 sess 16 trials
Rare difficult
2 sess 29 trials
MC06 — grand average — short block
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
ns
ns
n/a
ns
Common easy
2 sess 53 trials
Common difficult
2 sess 47 trials
Rare easy
2 sess 5 trials
Rare difficult
2 sess 19 trials
MC06 — grand average — long block
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
ns
ns
ns
ns
Common easy
2 sess 57 trials
Common difficult
2 sess 46 trials
Rare easy
2 sess 11 trials
Rare difficult
2 sess 10 trials

Block versus matched non-naive neutral P(Choose Long-Associated Response) — Four psychometric bins (outer bins define easy/hard)
Pooled: Fisher exact; grand average: paired Wilcoxon

All subjects — pooled — all blocks
All subjects — pooled — short block
All subjects — pooled — long block
All subjects — grand average — all blocks
All subjects — grand average — short block
All subjects — grand average — long block
MC06 — pooled — all blocks
MC06 — pooled — short block
MC06 — pooled — long block
MC06 — grand average — all blocks
MC06 — grand average — short block
MC06 — grand average — long block
Matched neutral
Block
P(Choose Long-Associated Response)
Common easy
Common difficult
Rare easy
Rare difficult
2 sess
ns
*
**

Bias distributions — Four psychometric bins (outer bins define easy/hard): median + IQR and two-sided tests versus zero
All visual summaries use median + IQR; Shapiro p < 0.01 selects Wilcoxon, otherwise one-sample t-test

All subjects — pooled — all blocks

All subjects — pooled — short block

All subjects — pooled — long block

All subjects — grand average — all blocks

All subjects — grand average — short block

All subjects — grand average — long block

MC06 — pooled — all blocks

MC06 — pooled — short block

MC06 — pooled — long block

MC06 — grand average — all blocks

MC06 — grand average — short block

MC06 — grand average — long block

Bin-resolved psychometric choice bias: block minus matched non-naive neutral

All subjects — pooled — all blocks
All subjects — pooled — short block
All subjects — pooled — long block
All subjects — grand average — all blocks
All subjects — grand average — short block
All subjects — grand average — long block
MC06 — pooled — all blocks
MC06 — pooled — short block
MC06 — pooled — long block
MC06 — grand average — all blocks
MC06 — grand average — short block
MC06 — grand average — long block
Common — Short ISI
Common — Long ISI
Rare — Short ISI
Rare — Long ISI
Choice bias: ΔP(Long-associated response) (block − matched neutral)
ISI (s)
0.5
1.0
1.5
2.0
-0.4
-0.2
0.0
0.2
0.4
0.6
0.8
1.0